THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL MCQ TEST SET 3

Full Syllabus (Chapters 1–11) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. When a student removes one bulb from a string of decorative bulbs, all the bulbs go off. This shows that the circuit has: 1

(a) many separate paths **(b)** no cell
(c) a single path for the current **(d)** no wire

2. A metal M does not displace copper from CuSO4 but copper displaces it from M's salt solution. M is: 1

(a) a non-metal **(b)** more reactive than copper
(c) less reactive than copper **(d)** as reactive as copper

3. A student says that a burning candle makes the wax vanish. The correct statement is: 1

(a) the wax changes into new substances such as carbon dioxide and water vapour
(b) the wax is destroyed
(c) the wax turns into the wick
(d) the wax only melts

4. A nurse says that the thyroid gland needs iodine. A diet lacking it may lead to: 1

(a) swelling of the neck **(b)** swelling of the eye
(c) swelling of the arm **(d)** swelling of the foot

5. A student places a thermometer in sunlight and another in shade. The one in sunlight reads higher due to: 1

(a) radiation **(b)** evaporation **(c)** conduction **(d)** convection

6. In 50 s he covers: 1

(a) 400 m **(b)** 20 m **(c)** 200 m **(d)** 100 m

7. The villi are absent in the intestine. The consequence is: 1

(a) no breathing **(b)** poor absorption of digested food
(c) no digestion **(d)** no excretion

8. In an experiment, aquatic plant produces 10 bubbles per minute at 30 cm from a lamp and 40 at 10 cm. This shows photosynthesis rate: 1

(a) increases with light intensity **(b)** decreases with light
(c) is independent of light **(d)** stops at night

9. A girl walks away from a mirror at 2 m/s for 3 s. How far has the distance between her and her image increased? 1

(a) 3 m **(b)** 12 m **(c)** 6 m **(d)** 2 m

10. If the Earth were upright (no tilt), the Sun at noon on the equator would be: 1

(a) overhead only in June **(b)** overhead twice
(c) overhead all year **(d)** never overhead

11. A student mixes equal volumes of acid and base. The mixture still turns blue litmus red. This suggests: 1

(a) no reaction
(b) the acid was stronger or more concentrated
(c) the mixture is neutral
(d) the base was stronger

12. If the bulb in the tester does not glow with iron nail, it could be because: 1

(a) nail is a plastic **(b)** iron is an insulator
(c) cell is stronger **(d)** the nail is coated with paint or rust

13. The mass of the iron nail after being kept in copper sulphate solution for long: 1

(a) becomes zero **(b)** increases **(c)** stays same **(d)** decreases

14. During the burning of a candle, heat and light are given out. This suggests: 1

(a) a reversible change **(b)** no change

(c) a chemical change **(d)** a physical change

15. If a person's pancreas stops making insulin, blood sugar will: 1

(a) rise **(b)** stay same

(c) become acidic always **(d)** fall to zero

16. In deserts, nights are cold and days are hot because sand: 1

(a) reflects all **(b)** heats slowly

(c) heats and cools quickly **(d)** stores heat

17. A boat sails 24 km in 3 h with a steady speed. In 5 h it travels: 1

(a) 8 km **(b)** 15 km **(c)** 120 km **(d)** 40 km

18. A blood sample has 5 million RBC per mm^3. In 2 mm^3 the number is: 1

(a) 7 million **(b)** 5 million **(c)** 2.5 million **(d)** 10 million

19. Which observation shows that a plant is making more food than it uses? 1

(a) it stops growing **(b)** it turns pale

(c) it gains mass and grows **(d)** it loses mass

20. A 20 cm object is placed 40 cm from a lamp and its shadow falls on a screen at 100 cm from the lamp. The shadow height is: 1

(a) 50 cm **(b)** 100 cm **(c)** 40 cm **(d)** 20 cm

21. During a lunar eclipse the Moon takes 1 hour to cross the Earth's umbra. The umbra's width relative to the Moon's diameter is about: 1

(a) about 2.5 times **(b)** half **(c)** equal **(d)** 10 times

22. A cook adds a pinch of baking soda to lemon-flavoured drink and it fizzes. This shows: 1

(a) no reaction **(b)** sugar reaction

(c) acid-carbonate reaction **(d)** salt reaction

23. A circuit has a cell, a switch, a bulb and a fuse in one loop. If the fuse melts, the bulb: 1

(a) glows brighter **(b)** glows dimmer **(c)** stays the same **(d)** goes off

24. In the situation above, R and S are ___ than hydrogen. 1

(a) unreactive gases **(b)** more reactive **(c)** equal **(d)** less reactive

25. Why does the silver spoon get black while stainless steel spoons remain bright? 1

(a) silver reacts with sulphur in air **(b)** silver is a non-metal

(c) steel does not **(d)** steel is a gas

26. A girl's period is irregular during the first year. This is: 1

(a) common as cycles settle **(b)** a disease

(c) rare **(d)** a danger

27. A chimney is made tall so that: 1

(a) rain cannot enter **(b)** smoke is hidden

(c) bricks cool **(d)** hot gases rise and draw fresh air

28. Timing 20 oscillations in 40.0 s reduces the percentage error to about: 1

(a) 0.25 per cent **(b)** 5 per cent **(c)** 50 per cent **(d)** 2.5 per cent

29. Which of the following explains the red colour of blood? 1

(a) insulin **(b)** the iron-containing haemoglobin

(c) bile **(d)** chlorophyll

30. Bread mould is a: 1

(a) saprotroph **(b)** insectivore **(c)** parasite **(d)** autotroph

31. Which of these helps us to see a non-luminous book? 1

(a) light from a lamp reflecting from the book

(b) the shadow of the book

(c) the book giving its own light

(d) light passing through the book

32. A student thinks 'the dark side of the Moon is always dark'. This is: 1

(a) correct on new moon only **(b)** correct
(c) correct on full moon only **(d)** incorrect

33. Which of the following reaction shows a salt being formed? 1

(a) $2\,H_2 + O_2 \longrightarrow 2\,H_2O$ **(b)** $HCl + NaOH \longrightarrow NaCl + H_2O$ **(c)** $2\,Mg + O_2 \longrightarrow 2\,MgO$ **(d)** $C + O_2 \longrightarrow CO_2$

34. Which statement about a fuse wire is correct? 1

(a) its melting point is higher
(b) it never melts
(c) its melting point is lower than that of the connecting wires
(d) it is made of rubber

35. A student passes the gas formed on burning a substance through lime water and it turns milky. The substance contained: 1

(a) iron **(b)** sulphur **(c)** sodium **(d)** carbon

36. Which observation would show a chemical change in a mixture of sand and iron filings on heating with sulphur? 1

(a) a new black solid not attracted by a magnet
(b) sand settles
(c) colour same
(d) iron attracted by magnet

37. Which of the following hormone pairs is correctly matched with the gland? 1

(a) adrenaline – thyroid **(b)** testosterone – adrenal
(c) thyroxine – pancreas **(d)** insulin – pancreas

38. A wall painted white keeps the room cooler than a wall painted black because: 1

(a) white absorbs **(b)** white conducts
(c) black reflects **(d)** white reflects most radiation

39. A car goes uphill at 20 km/h and downhill at 60 km/h over the same distance. The average speed is: 1

(a) 30 km/h **(b)** 40 km/h **(c)** 20 km/h **(d)** 80 km/h

40. If the diaphragm fails to contract: 1

(a) kidneys stop **(b)** inhalation is affected
(c) heart stops **(d)** digestion stops

41. **Assertion (A):** The Moon is a non-luminous object. 1
Reason (R): The Sun is a luminous object.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

42. **Assertion (A):** The seasons are caused by the changing distance of the Earth from the Sun. 1
Reason (R): The axis of the Earth is tilted.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

43. **Assertion (A):** Neutralisation is the reaction between an acid and a base to form salt and water. 1
Reason (R): Heat is released during a neutralisation reaction.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

44. **Assertion (A):** Electric wires are covered with plastic or rubber. 1
Reason (R): Copper is a good conductor of electricity.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

45. **Assertion (A):** An iron nail dipped in copper sulphate solution gets a reddish-brown coating. 1
Reason (R): Iron is more reactive than copper and displaces it from its solution.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** Burning of a candle involves both a physical and a chemical change. 1
Reason (R): The wax melts (physical) and the wax vapour burns to form new substances (chemical).
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** Testosterone is secreted by the testes. 1
Reason (R): Oestrogen is secreted by the ovaries.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** Convection takes place in liquids and gases. 1
Reason (R): Conduction takes place in solids.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** A car that travels 90 km in 2 hours has a speed of 45 km/h. 1
Reason (R): Speed is equal to distance multiplied by time.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** Arteries carry blood towards the heart. 1
Reason (R): Veins carry blood towards the heart.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**